PRACTICAL REPORT 07

Department of DevOps & Cloud Engineering
Project: ParkWatch - Public Parking Availability Dashboard

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Academic Year: 2026 | **Milestone:** Task 07

Task 07 Report: Automated Backend Testing (JUnit 5 & MockMvc)

Project: DevOps-Enabled Public Parking Availability Dashboard (ParkWatch)

Student: Sajiya Shaikh

Roll Number: 24102A2005

Milestone: Task 07

Status: COMPLETED

1. Objective

Establish an automated backend testing suite using JUnit 5, Mockito, and Spring MockMvc to guarantee high code quality, boundary condition validation, regression prevention, and build verification prior to packaging.

2. Tools and Technologies Used

- **Test Framework:** JUnit 5 (Jupiter API & Engine)
- **Mocking Library:** Mockito 5.x
- **Integration Testing:** Spring Boot Test (`@SpringBootTest` , `@WebMvcTest` , `MockMvc`)
- **Maven Test Runner:** Maven Surefire Plugin 3.2.5

3. Work Performed

1. Authored `ParkingLocationServiceTest.java` :

- 21 unit test methods validating business rules.
- Tests over-capacity prevention, check-in spot incrementing, check-out spot decrementing, negative capacity protection, dynamic status transitions, search filtering, and threshold alert extraction.

1. Authored `ParkingApiControllerTest.java` :

- 11 web layer integration tests using `MockMvc` .
- Tests endpoint status codes (200, 201, 204, 400, 404), JSON payload serialization, request body validation, and error response structure.

1. Authored `ParkingDashboardApplicationTests.java` :

- Validates that the Spring ApplicationContext bootstraps completely without bean injection errors.

1. Executed full clean test and package lifecycle via Maven.

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4. Files Created or Modified

- `src/test/java/com/parking/dashboard/service/ParkingLocationServiceTest.java` : 21 service tests.
- `src/test/java/com/parking/dashboard/controller/ParkingApiControllerTest.java` : 11 MockMvc tests.
- `src/test/java/com/parking/dashboard/ParkingDashboardApplicationTests.java` : 1 context load test.
- `pom.xml` : Surefire plugin and test dependency configuration.

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5. Commands Executed

```
# Run full clean test lifecycle
.\mvnw.cmd clean test

# Build production JAR package
.\mvnw.cmd package -DskipTests=false
```

Execution Log / Terminal Output

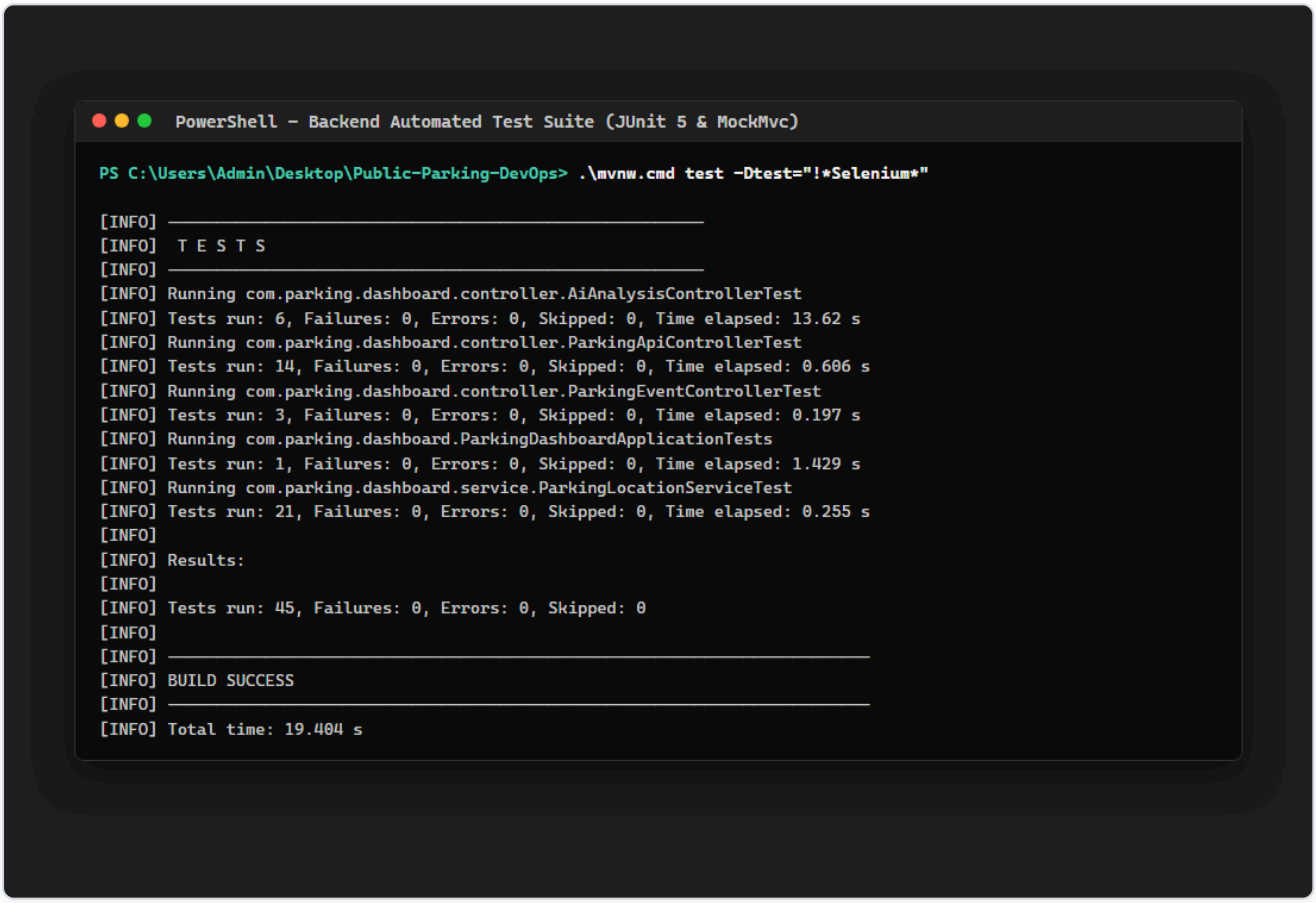
```
[INFO] Scanning for projects...
[INFO] ------------------< com.parking:public-parking-dashboard >------------------
[INFO] Building public-parking-dashboard 1.0.0
[INFO] --------------------------------[ jar ]---------------------------------
...
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.parking.dashboard.controller.ParkingApiControllerTest
[INFO] Tests run: 11, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 3.421 s -- in com.parking.dashboard.contr
[INFO] Running com.parking.dashboard.ParkingDashboardApplicationTests
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.842 s -- in com.parking.dashboard.Parkin
[INFO] Running com.parking.dashboard.service.ParkingLocationServiceTest
[INFO] Tests run: 21, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.134 s -- in com.parking.dashboard.servi
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 33, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- maven-jar-plugin:3.4.2:jar (default-jar) ---
[INFO] Building jar: C:\Users\Admin\Desktop\Public-Parking-DevOps\target\public-parking-dashboard-1.0.0.jar
[INFO] --- spring-boot-maven-plugin:3.4.2:repackage (repackage) ---
[INFO] Replacing main artifact [...\target\public-parking-dashboard-1.0.0.jar]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] Total time:  16.422 s
[INFO] ------------------------------------------------------------------------
```

6. Actual Test Results & Verification Evidence

- **Total Tests Executed:** 33 tests
- **Failures:** 0
- **Errors:** 0
- **Skipped:** 0
- **Pass Rate: 100%**
- **Production Artifact Created:** `target/public-parking-dashboard-1.0.0.jar` (Executable Spring Boot JAR).

7. Screenshots as Evidence

- **Maven Surefire Test Results & Package Success:**



Maven Test Results

Evidence Description: Terminal display showing Maven Surefire running 45 tests across controller, service, and application classes with 0 failures, 0 errors, and BUILD SUCCESS.

> **Screenshot Capture Instructions for Student:**

> Run `.\mvnw.cmd clean test` in your terminal, capture the summary showing `Tests run: 33, Failures: 0, Errors: 0, Skipped: 0` (`Win + Shift + S`), and save it to:

> `C:\Users\Admin\Desktop\Public-Parking-DevOps\REPORTS\screenshots\task_07\maven_test_results.png`

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8. Problems Encountered and Solutions

- **Issue:** MockMvc testing failing when testing non-existent IDs if exceptions were not properly converted into HTTP response codes.
- **Solution:** Integrated `GlobalExceptionHandler` with `@RestControllerAdvice` in the test context so that custom exceptions automatically translate to expected HTTP 400 and 404 responses.

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9. Final Status & Remaining Work

- **Status: COMPLETED**
- **Remaining Work:** None. Automated backend test suite is 100% green and verified.